Notes and errata from an implementation of the Amato polygon-separation algorithm

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Abstract. We describe what is, to our knowledge, the first implementation of the $\Theta(n)$ polygon-separation algorithm of Amato [1], undertaken as part of the `geo` computational-geometry library. The implementation is validated by property-based testing against a brute-force oracle over randomly generated simple polygons. This process surfaced five configurations in which the decomposition step, as described in [1], is underspecified or fails to cover the pair realising the separation; each is perturbation-stable and arises in exact arithmetic. For each we give a minimal integer-coordinate counterexample, an analysis, and the resolution adopted, which are then collected into an amended statement of the subchain-construction step, with a proof that it assigns every visible pair to some subproblem. Most of the missing detail is present in the earlier technical report [2], whose axiomatic treatment of the separator sequence the 1994 paper compresses; two findings appear to go beyond both papers. We would welcome corrections to our reading.

1 Introduction

The separation $\sigma(P, Q)$ of two simple polygons – the minimum distance between their boundaries – can be computed in optimal $\Theta(n)$ sequential time by the decomposition algorithm of [1], which reduces the problem to linearly separable subproblems using a sequence of separating segments derived from a shortest path, and solves each subproblem with the closest-visible-vertex machinery of the technical report [2] (published in revised form as [3]; we cite the report, whose section numbering we follow). We have implemented the sequential algorithm in Rust for the `geo` library[1].

Nothing in these notes disputes the papers' central results. Our findings concern the decomposition step (DECOMPOSE, Step 1(b–d) and Step 2 of [1]), whose full treatment the 1994 paper compresses from [2]. Our experience is that this compression omits important detail: an implementer working from [1] alone will, on certain perturbation-stable inputs, compute an overestimate of $\sigma(P, Q)$. Most of the machinery needed to repair this is present in [2]; in two places we believe the configurations fall outside both papers.

Each finding was discovered by a property-based test harness comparing the implementation against a brute-force oracle on randomly generated simple polygons (Section 6). Each is presented with a minimal integer-coordinate counterexample, simplified from the harness's shrunk original and verified to reproduce the defect by reverting the corresponding repair. There is precedent for such findings in this line of work: [2] itself demonstrates that the two prior published algorithms for the closest-visible-vertex problem are incorrect on some instances – the subproblem decomposition of [4] "can produce subproblems whose solutions may not always be valid solutions for the original problem", and Lemma 2.2 of [5] does not hold as stated.

2 The separator sequence, as defined in the 1992 report

[2] (Section 4) defines $S(P, Q) = (l_0, \ldots, l_{m-1})$ as any sequence of segments satisfying:

[1] https://github.com/georust/geo

1. $l_i \cap l_{i+1} \neq \varnothing$ (consecutive separators intersect);
2. $l_i \cap l_j = \varnothing$ for $j \notin \{i-1, i, i+1\}$ (non-consecutive ones do not);
3. l_i does not intersect the interior of P or Q;
4. **both endpoints of l_i lie on the boundaries of P and/or Q**;
5. **each l_i is maximal**: if the top endpoint of l_i is on P and the top endpoint of l_{i+1} is on Q, no point of Q at or above l_{i+1}'s top is visible from l_i, and symmetrically.

[2] notes that (v) implies each l_i intersects the boundaries of both P and Q. The subchain-truncation rules of Step 2 are sound *given* these properties; they are theorems about sequences satisfying (i)–(v), not about arbitrary extended geodesics. [1] restates the consequence ("it is a simple matter to verify that each l_i intersects both P and Q") without the properties; its one-path construction establishes (i)–(iii) but not (iv) or (v) (Appendix A). That is the root of the findings in Section 3.

A second structural difference: the construction in [2] (Theorem 3) computes **two** shortest paths – L_p between the two P-contacts and L_q between the two Q-contacts – takes their common subpath, extends each segment **one-sidedly** beyond the shared path vertex, and determines which polygon an extension endpoint lies on **by provenance** ("by inspecting the (unextended) segments"), not by geometric search. The monotonicity of the cut points ("if $l_i^+ \in P$ and $l_j^+ \in P$, $i < j$, then $l_i^+ < l_j^+$") is proved from the two-path structure. [1] uses a single path between the P-contacts and extends in both directions, and none of those guarantees carry over.

The repairs described below share one safety principle, valid for the separation problem but **not** for the closest-visible-vertex problem (Section 5.2): enlarging a subchain is always sound, because every value a subproblem produces is a distance between boundary points and therefore at least σ, while the pair realising σ only gains coverage.

3 Findings

Each finding states the claim, a counterexample (as WKT, with a figure drawn from the same coordinates), an analysis, and the resolution adopted in our implementation. In the figures, supporting segments (bridges) are dashed, the geodesic is dotted, and separators are solid red.

3.1 A separator can miss one polygon entirely

Claim ([1], Section 3): every separator intersects both polygons; the cut points p_i^+ and q_i^+ always exist.

Counterexample:

```
P: POLYGON((0 2, 0 -2, 2 0, 0 2))
Q: POLYGON((-6 -2, -6 -6, -2 -5, -6 -2))
```

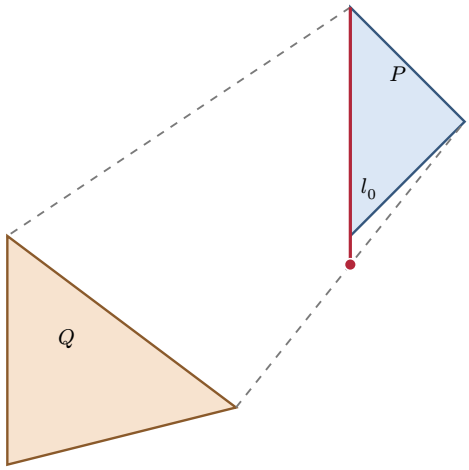


Figure 1: The separator l_0 is the extension of the first geodesic segment; it terminates on a bridge and does not intersect Q, so the cut q_0^+ does not exist.

Two triangles whose channel axis is diagonal (Figure 1). The geodesic in R runs along P's facing arc; the extension of its first segment leaves P and lands on a **bridge** (a supporting segment of R's boundary), touching Q nowhere. In exact arithmetic, $l_0 \cap Q = \varnothing$: no q_0^+ exists. The configuration is an open set – perturbing the vertices preserves it. The "otherwise" branch of Step 2 (end the subchain at the next separator's cut) then truncates the realising vertex of Q out of every subchain, and the computed separation exceeds the true value.

Analysis. This violates property (iv) of [2], which the one-path construction of [1] does not establish. R's boundary consists of the two facing arcs *and the two bridges*; nothing prevents an extension from landing on a bridge. We have not determined whether the two-path construction of [2] excludes bridge landings in its own setting, where the machinery is invoked only for polygons whose hulls intersect; in the [1] setting, where DECOMPOSE also serves hull-disjoint polygons, they occur.

Resolution. The truncation branches that assume the missing cut exists are replaced by the facing-arc sentinel (Section 3.3), and each extension records the boundary edge it hit, so a bridge landing is recognised exactly rather than searched for geometrically.

3.2 The orientation rule contradicts the path-order conventions

Claim ([1]): the bottom endpoint b_i of l_i is "the endpoint closest to l_{i-1}", the top t_i the endpoint closest to l_{i+1}.

The sentinel conventions ($p_m^+ = p_t$, and the last separator's cut landing at the top contact) presuppose that this orientation agrees with the direction of the path.

Counterexample: the triangle pair of Figure 1. The backward extension of the second geodesic segment overshoots so far that the extended segment's far endpoint is *more distant* from l_0 than its top endpoint: the closest-endpoint rule labels the endpoints in the opposite sense to path order, p_1^+ becomes the wrong intersection, and the last subchain degenerates.

Analysis. [2] does not use a distance rule. It defines the anchors as the *shared path vertices* ($l_i^- = l_{i-1} \cap l_i$, $l_i^+ = l_i \cap l_{i+1}$) and extends one-sidedly, so orientation is path order by construction. The distance formulation of [1] is an operationalisation that agrees with path order in the generic figure and diverges when an extension overshoots.

Resolution. Orientation is taken from path order, matching the definition in [2] rather than the operationalisation in [1].

3.3 The truncation branches require property (v)

Claim ([1], Step 2): the subchain of X for separator i ends at i's own cut when $t_i \in X$ **or** $\boldsymbol{i = m - 1}$, and at separator $i + 1$'s cut otherwise.

Counterexamples: Section 3.1 breaks the "otherwise" branch (the next cut does not exist, or truncates the realising vertex). The "$i = m - 1$" branch fails on:

```
P: POLYGON((2 -4, -1 3, 7 12, -4 8, -4 -4, 2 -4))
Q: POLYGON((12 4, 6 8, 4 4, 8 0, 12 4))
```

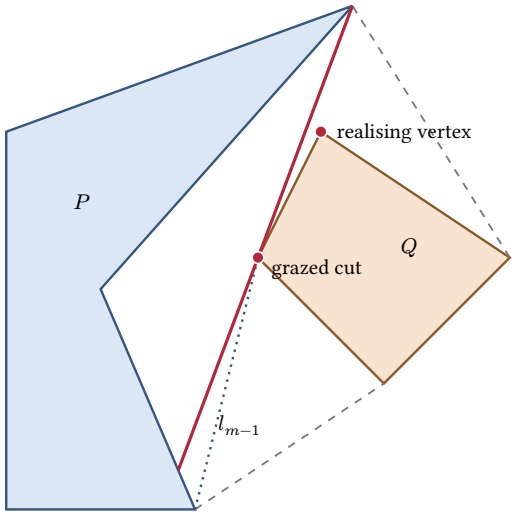


Figure 2: The last separator is anchored on P at both ends and passes through a vertex of Q without terminating into Q's wall; the realising vertex lies beyond the grazed cut, which shields nothing.

The last separator is anchored on P at both ends (Figure 2) – the geodesic turned at a vertex of Q, so the extended segment *grazes* Q at that vertex without terminating into Q's wall. Its "own cut" on Q is the grazed vertex; it shields nothing, and the realising pair (assigned to this subproblem by the coverage lemma) lies beyond it.

Analysis. In [2] both truncations are justified by property (v), which the grazing configuration violates: the separator is not maximal with respect to Q. Geodesics generically turn at reflex vertices of the channel boundary, and the channel boundary includes both polygons, so separators grazing the opposite polygon are not exotic.

Resolution. A subchain end stops at separator i's own cut exactly when i's top endpoint lands on that polygon (established by hit-edge provenance); a subchain start begins at separator $i - 1$'s cut exactly when $i - 1$'s bottom endpoint lands on that polygon. In every other situation the subchain extends to the facing-arc sentinel. These are precisely the situations in which the [2] shielding arguments go through; when the endpoint condition holds, the cut exists, since the endpoint itself lies in $l_i \cap P$.

3.4 Fixed-direction subchain scans can wrap

Claim ([1]): subchains are fixed-direction scans between consecutive cut points; "no point of P or Q appears in more than four subproblems", so the aggregate subproblem size is $O(n)$. ([2] says five, not four, for its closely related construction.)

Counterexample:

```
P: POLYGON((0 0, -1 -2, 2 -1, 0 0))
Q: POLYGON((-8 -4, -8 -8, 4 -6, -8 -4))
```

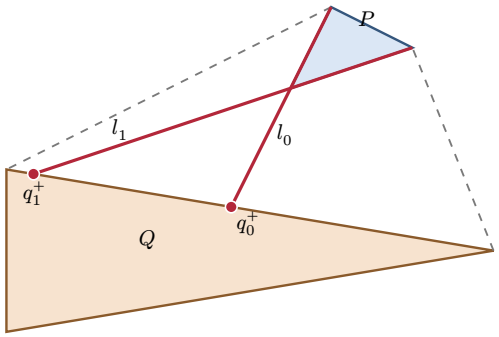


Figure 3: l_1's backward extension lands before l_0's cut in boundary order (q_1^+ precedes q_0^+), so the fixed-direction subchain scan between the cuts wraps around Q.

A later separator cuts Q *below* an earlier one (Figure 3): the backward extension of the second geodesic segment lands far down the opposite wall, before the earlier separator's cut in boundary order. The pair realising σ touches l_0, so the coverage lemma asserts it lies in the first subproblem, whose Q-subchain ends at q_0^+; but while the pair's Q-point lies below q_0^+ in the height order along the separator, which the lemma's proof uses, it lies beyond q_0^+ in the boundary order the subchain uses. The two orders come apart, and only a scan continuing past q_0^+ – wrapping around the polygon – reaches the point.

Analysis. [2] proves the cut points ascend, but from the two-path construction; under the one-path construction of [1], descending cuts occur. When they do, either the subchain wraps (and the four-per-vertex aggregate claim no longer follows) or coverage is lost. The slippage is between two orders the construction treats as interchangeable: position along a separator, which the coverage lemma's proof uses, and position along the boundary, which the subchains use. This appears to be a gap in the complexity argument of [1] for configurations its construction can produce.

Resolution. An earlier version of the implementation detected an anchor cut lying outside the computed slice of the facing arc and substituted the full polygon ring for that subchain, a sound superset. The coverage proof (Section A.4) shows the override is never needed: the pairs it concerns are covered by the neighbouring subproblems, and the slice between the cuts is now kept as it is. The $O(n)$ aggregate bound is deliberately given up in these cases; restoring it is a matter for adopting the [2] two-path construction, whose monotonicity argument rules the wraps out.

3.5 The single-contact ("pinched") channel

Claim ([1], Step 1(a)): the non-containing case yields a simple polygon R bounded by two facing portions and two supporting segments, with distinct path endpoints $p_b \neq p_t$.

Counterexample:

```
P: POLYGON((3 2, 0 -4, 4 2, 3 2))
Q: POLYGON((0 8, 2 3, 8 4, 0 8))
```

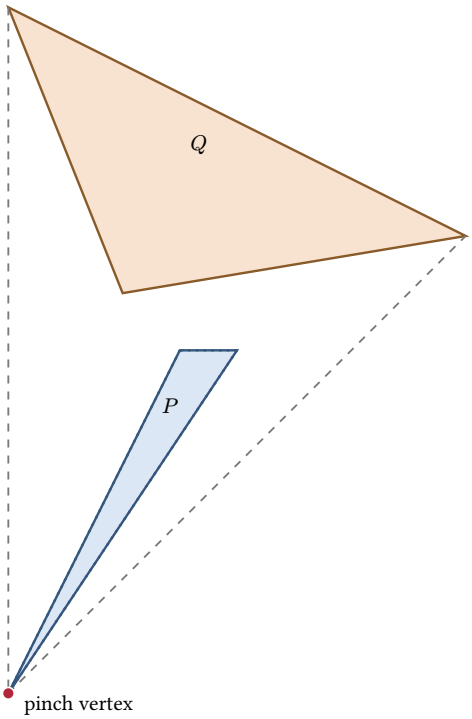


Figure 4: Both supporting segments (dashed) share P's single hull contact: the channel is pinched there, and the shortest path (dotted) wraps around P between the two instances of the pinch vertex.

P touches $\mathrm{CH}(P \cup Q)$ at a single vertex (Figure 4). Both supporting segments share that P-contact, P's facing portion is its entire boundary, R is weakly simple (pinched at the shared vertex, which appears twice in its ring), and $p_b = p_t$. The shortest path runs between the two instances of the pinch vertex, around P – structurally the containing case's doubled cut vertex, inside the non-containing case's framing. The linear bottom-to-top subchain rules do not apply: the separator sequence is cyclic, as in the containing case's "all arithmetic is modulo m".

Analysis. Neither paper discusses the configuration. It is not excluded by the general-position assumption of [2] (no three of the counterexample's vertices are collinear).

Resolution. Pinched channels are detected exactly (shared bridge contact) and take full-ring subchains for every separator – sound, with the containing-case machinery as the eventual proper treatment. The shortest-path and visibility machinery is index-based, so the two instances of the pinch vertex are distinct vertices, which is what forces paths to route around the pinched-in polygon.

4 An amended construction

We collect the resolutions of Section 3 into an amended statement of the affected steps, in the style of Algorithm 2 of [1]. Only the diverging parts are restated; the pointers give the finding motivating each change. The statement is correctness-first: it preserves the coverage of every visible pair, but weakens the complexity guarantee. A sentinel-extended subchain runs to the end of its facing portion, and by Lemma A4 of Appendix A the geodesic segment inside each separator ends on one wall, so at each end of every subproblem one of the two subchains runs to its sentinel. Whenever the geodesic alternates between the walls $\Theta(n)$ times, the aggregate subproblem size is therefore $\Theta(n^2)$, against $\Theta(n)$ in [1]: the quadratic bound is tight, and it is the typical behaviour rather than a worst case. A sentinel-extended subchain need not lie on one side of its separator either, so such subproblems are not linearly separable and an

implementation must solve them by other means. Bounding the sentinel side by a later cut is exactly what property (v) supplies, and securing it is the role of the two-path construction of [2] (Section 3.4). Coverage of every visible pair by the amended rules is proved in Appendix A and confirmed empirically (Section 6).

Amended DECOMPOSE(P, Q), Steps 1(c)–2, non-containing case

Step 1(c). *Extend the segments of the shortest path to R's boundary.* One ray per direction; an endpoint whose ray leaves R immediately is not extended. Record, for each extension, the feature of R's boundary on which it lands: an edge of P', an edge of Q', or a bridge (Section 3.1). Orient each extended segment by path order: b_i is the endpoint on the side of the path's start, t_i the other (Section 3.2).

Step 1(d). *Remove redundant segments.* Unchanged.

Step 2. *Construct the subproblems.* For each $l_i \in S(P, Q)$ let p_i^+ be the highest point of intersection of l_i with P along the orientation of Step 1(c), when one exists (Section 3.1), and let the subchain run along the facing portion P'. Define $P_i = P_{s_i, e_i}$, where

$$e_i = \begin{cases} p_i^+ \text{ if } t_i \in P \\ p_t \quad \text{otherwise,} \end{cases} \qquad s_i = \begin{cases} p_{i-1}^+ \text{ if } i > 0 \text{ and } b_{i-1} \in P \\ p_b \quad \text{otherwise} \end{cases}$$

except that if the channel is pinched, then $P_i = \partial P$ for every i (Section 3.5).

Q_i is defined symmetrically. As in [1], $\sigma(P, Q) = \min_{0 \le i < m} \sigma(P_i, Q_i)$.

For comparison, Step 2 of [1] reads

$$P_i = \begin{cases} P_{p_{i-1}^+, p_i^+} & \text{if } t_i \in P \text{ or } i = m - 1 \\ P_{p_{i-1}^+, p_{i+1}^+} & \text{otherwise} \end{cases}$$

with the subchain taken as a fixed-direction scan of the whole polygon. The amendments are: the removal of the "$i = m - 1$" disjunct (Section 3.3); the replacement of the neighbouring cut p_{i+1}^+ by the sentinel p_t (Section 3.1, Section 3.3); the added start condition with its sentinel p_b (Section 3.3); and the restriction of subchains to the facing portions, with the pinch override (Section 3.5) replacing the wrapping scans; the descending cuts of Section 3.4 need no rule of their own (Section A.4).

5 Remarks

5.1 General position

[2] assumes no three vertices are collinear "for ease of exposition", stating the algorithms remain valid without alteration. The counterexamples above satisfy general position, so the assumption does not account for them.

5.2 Separation versus closest visible vertices

[1] correctly observes that $\sigma(P_i, Q_i) \ge \sigma(P, Q)$ holds for arbitrary subchains, which is why the separation problem tolerates enlarged subchains: this one-sided safety underpins the sentinel and whole-boundary resolutions above. The closest-visible-vertex problem does **not** enjoy it: a subproblem can contain a vertex pair that is closer than $\mathrm{CVV}(P, Q)$ but not visible, and returning it is an error. [2] devotes Lemma 11 and Figure 9(b) to exactly this hazard and prunes the affected subchains. Any adaptation of the repairs above to CVV must add that machinery; enlarging subchains is not sound there.

5.3 Floating-point transfer

Three further failure classes arose in transferring the real-RAM algorithm to IEEE-754 doubles; they are implementation matters, not errata. Exact segment-intersection tests miss computed extension endpoints by an ulp, repaired by carrying exact provenance instead of re-deriving contacts geometrically, an approach anticipated by the instruction in [2] to inspect the unextended segments. A spurious hit of the extension ray on an edge incident to the vertex it passes through, at parameter $1 + \varepsilon$, masks the true extension through a reflex vertex, repaired by excluding those edges from the scan by index. Finally, exactness of Shewchuk-style robust predicates fails when orientation products underflow (magnitudes below roughly 10^{-323}), which affected our test oracle rather than the algorithm.

6 Verification methodology

The implementation is validated by property-based testing. Input pairs are star-shaped simple polygons with 3–12 vertices, generated by drawing bounded angular gaps (normalised so that every gap is below 180°, which guarantees simplicity) and varying radii; pairs whose boundaries intersect are discarded. The oracle is the $O(n^2)$ minimum over all boundary point–segment distances. The property asserts agreement of the implementation with the oracle within a scale-aware tolerance (relative 8ε, with an absolute floor of 32ε times the coordinate magnitude, where $\varepsilon = 2^{-52}$ is the binary64 machine epsilon: both sides round at that scale, and at knife-edge inputs exact pruning decisions interact with rounded distance evaluations on either side of the true value). At the time of writing the property holds over 300,000 generated pairs, with the individual pipeline stages covered by their own unit fixtures and properties.

The counterexamples in this document are integer-coordinate simplifications of the harness's shrunk originals. Each was verified to reproduce its defect by reverting the corresponding resolution and observing the oracle mismatch, and to pass under the unmodified implementation; all four are in general position. Both the simplified and the original shrunk forms are retained as regression fixtures in the implementation's test suite.

7 Closing remarks

We hope these notes are useful as a record of what a correctness-first implementation of the separation algorithm encounters, three decades on. We would welcome corrections to our reading of either paper – in particular, whether the two-path construction of [2] was intended to establish properties (iv) and (v) in the hull-disjoint setting served by DECOMPOSE, and whether the wrapping subchain scans of Section 3.4 were considered in the aggregate-size analysis. Appendix A proves that the amended rules cover every visible pair; the linear aggregate bound is the question that remains. The implementation, including the property harness and all fixtures, is being prepared for inclusion in the `geo` library.

A Coverage of the amended construction

We prove that the amended Step 2 of Section 4 assigns every visible pair to some subproblem, in the non-containing case. The pinch override makes every subchain a whole boundary, so coverage there is immediate; we assume throughout that the channel is not pinched. The argument is elementary plane topology together with one metric fact, the uniqueness of geodesics in a simple polygon. It does not use properties (ii), (iv) or (v) of [2] as hypotheses; property (ii) is shown to follow from Step 1(d), and the places where (iv) and (v) fail are exactly the places where the sentinels of Section 4 take over.

A.1 Setting and notation

P and Q are simple polygons with disjoint boundaries, in the non-containing case, with bridges $p_b q_b$ (bottom) and $q_t p_t$ (top) meeting $P \cup Q$ only at their contacts. The facing arc P' is the arc of ∂P from p_b to p_t that bounds R; Q' is the arc of ∂Q from q_b to q_t that bounds R. Walking ∂R with R on the left visits

p_b, the bottom bridge, Q' from q_b to q_t, the top bridge, then P' from p_t back to p_b. *Arc order* on P' runs from p_b to p_t, on Q' from q_b to q_t; "above" and "below" refer to arc order on the arc in question, never to a direction in the plane. Facing up the channel, P' is the left wall and Q' the right wall. We call P' and Q' the two *walls* and say that points of P' and Q' lie on *opposite* walls.

$\pi = (v_0 = p_b, v_1, ..., v_k = p_t)$ is the geodesic in $\mathrm{cl}(R)$ from p_b to p_t, listed with every boundary vertex it passes through, so that consecutive vertices may be collinear. Each v_j lies on a wall; every open segment (v_j, v_{j+1}) lies in the interior of R or along an edge of R. π is a simple arc. Its interior vertices are reflex vertices of R or grazed vertices; at a P' vertex where it turns, it turns left, at a Q' vertex right (it wraps the polygon whose vertex it passes).

For a path segment $s_j = v_j v_{j+1}$, the *extended segment* l_j is s_j prolonged in both directions while the prolongation stays in the interior of R, up to the first boundary point in each direction; a direction in which the prolongation leaves R at once is not extended. l_j is oriented by path order: b_j is its endpoint on the side of v_j, t_j on the side of v_{j+1}. The kept sequence $S(P, Q) = (l_0, ..., l_{m-1})$ is the output of Step 1(d); $w(i)$ denotes the path window of l_i, so $w(0) = 0 < w(1) < ... < w(m-1) = k-1$ and l_i is the extension of $s_{w(i)}$.

The *run* of l_i is $l_i \cap \pi$; by Lemma A2 below it is the maximal collinear stretch of π through $s_{w(i)}$, a segment from a path vertex u_i^- to a later path vertex u_i^+. The *lower arm* of l_i is $[b_i, u_i^-)$ and the *upper arm* is $(u_i^+, t_i]$; either may be empty. x_i denotes a point of $l_i \cap l_{i+1}$.

For $X \in \{P, Q\}$ we write x_i^+ for the highest point of $l_i \cap X'$ along the orientation of l_i, when the intersection is non-empty. The subchains are those of Section 4: X_i is the arc of X' from s_i^X to e_i^X where $e_i^X = x_i^+$ if $t_i \in X'$ and $e_i^X = x_t$ otherwise, and $s_i^X = x_{i-1}^+$ if $i > 0$ and $b_{i-1} \in X'$, $s_i^X = x_b$ otherwise; the whole-boundary override once used for Section 3.4 is discussed in Section A.4 and is not needed. Landing on a contact vertex counts as landing on that wall.

A pair (p, q), $p \in \partial P$, $q \in \partial Q$, is *visible* if the open segment (p, q) meets neither boundary. Its *sight* is the closed segment $[p, q]$.

A.2 Preliminaries

We use one topological tool repeatedly. Let $K \subset \mathrm{cl}(R)$ be closed and connected, and let γ be a path in $\mathrm{cl}(R)$ avoiding K whose endpoints y, y' lie on ∂R. Then y and y' lie in the same component of $\partial R \setminus K$. (If not, choose $k_1, k_2 \in K \cap \partial R$ separating y from y' on the circle ∂R; a simple sub-arc of γ is a cross-cut of the disc $\mathrm{cl}(R)$ that separates k_1 from k_2, contradicting the connectedness of K.) We refer to this as the *separation principle*. Two special cases: a segment in $\mathrm{cl}(R)$ with endpoints on ∂R separates the two open arcs it cuts ∂R into; and a sight from P' to Q' meets every connected set that contains p_b and p_t, since $P' \setminus \{p_b, p_t\}$ and Q' lie in different components of $\partial R \setminus \{p_b, p_t\}$.

Lemma A1 (confinement). *For every visible pair, $p \in P'$, $q \in Q'$, and the sight lies in* $\mathrm{cl}(R)$. *If the open sight meets a bridge, the pair is (p_b, q_b) or (p_t, q_t).*

Proof. Let $H = \mathrm{CH}(P \cup Q)$. Its boundary consists of the two bridges and two monochromatic chains, the hull P-chain from p_t to p_b and the hull Q-chain from q_b to q_t. A Jordan curve inside a closed convex disc meets the boundary in the same cyclic order as the curve itself, so the hull vertices of P occur on ∂P in hull order and $P' \cap \partial H = \{p_b, p_t\}$; P' cannot itself be a hull edge, since P and H traverse it in opposite senses. No point of Q lies on the hull P-chain: a point of Q interior to a hull edge uu' with $u, u' \in P$ would put all of Q (connected, disjoint from ∂P) inside the pocket bounded by uu' and the arc of ∂P between u and u', but q_b lies on a bridge, outside that pocket. Hence P' and Q' are disjoint chords of the disc H with non-interleaved endpoints, and H is the union of $\mathrm{cl}(R)$ with the regions K_P (bounded by the hull P-chain and P') and K_Q, with disjoint interiors, $\partial Q \cap K_P = \varnothing$ and $\partial P \cap K_Q = \varnothing$.

The sight lies in H by convexity. If $p \in \operatorname{int}(K_P)$, or p lies on the hull P-chain, the sight must leave K_P to reach $q \notin K_P$; near a hull-chain point other than p_b, p_t the sets K_P and H coincide, so the exit point lies on P' or is p_b or p_t, in the open sight, contradicting visibility. Hence $p \in P'$, and symmetrically $q \in Q'$. The same exit argument keeps the open sight out of $\operatorname{int}(K_P)$ and $\operatorname{int}(K_Q)$, so it lies in $\operatorname{cl}(R)$. A segment inside a convex set that meets the boundary at a relative-interior point lies in a supporting line, so an open sight meeting a bridge lies along that bridge, and its endpoints are the bridge's contacts. □

Lemma A2 (straightness). *For every j, $l_j \cap \pi$ is the maximal collinear stretch of π containing s_j. In particular an arm of l_j meets π only at its own path vertex, and two extended segments meet either at a shared path vertex or at a point of an arm of each.*

Proof. Let $y \in l_j \cap \pi$ and let v be the endpoint of s_j on the side of y. The segment $[v, y] \subset l_j$ lies in $\operatorname{cl}(R)$, so it is a shortest path from v to y; the sub-path of π between v and y is also a shortest path; geodesics in a simple polygon are unique, so the two coincide and π is straight from v to y. This is the observation of [2] (proof of Theorem 3) that no extended segment meets an unextended one except at endpoints. □

The same uniqueness argument shows that a sight, being the geodesic between its endpoints, meets π in a connected set: a point, or a collinear segment.

Lemma A3 (ordering). *The P' vertices of π occur in increasing arc order along P'; likewise the Q' vertices along Q'.*

Proof. Suppose not. Among pairs v_a, v_c of P' vertices of π with $a < c$ and v_c below v_a, choose one whose arc interval $[v_c, v_a]$ contains no other P' vertex of π; one exists, since any third vertex inside the interval yields a violating pair with a smaller interval. Then $J = \pi[v_a, v_c] \cup P'[v_c, v_a]$ is a simple closed curve; let D be the region it bounds. The sub-arcs $\sigma_1 = \pi[p_b, v_a]$ and $\sigma_2 = \pi[v_c, p_t]$ avoid $\operatorname{int}(D)$: to enter it they would have to cross J, which is impossible along $\pi[v_a, v_c]$ (simplicity) and along $P'(v_c, v_a)$ (no P' vertex of π there). At v_c the interior of R is split by the arriving segment of π into the wedge on the side of D and the wedge adjacent to the P' edge running downward from v_c; σ_2 leaves into the latter. Hence a point $y \in P'$ just below v_c and p_t are joined by a path avoiding $K = \sigma_1 \cup \pi[v_a, v_c]$. But K is connected, contains p_b and v_c, and y and p_t lie on different arcs of $\partial R \setminus \{p_b, v_c\}$, contradicting the separation principle. □

The lemma is purely topological: it holds for any simple arc from p_b to p_t in $\operatorname{cl}(R)$.

Pockets. By Lemma A3 the components of $\operatorname{cl}(R) \setminus \pi$ are of two kinds (Figure A.1). A *P-pocket* $\Pi(v_a, v_c)$ is bounded by the arc $P'[v_a, v_c]$ between two consecutive P' vertices of π and by $\pi[v_a, v_c]$, whose interior vertices all lie on Q'. A *Q-pocket* is defined symmetrically, except that the bottom Q-pocket is bounded by $\pi[p_b, v]$, the bottom bridge and $Q'[q_b, v]$, where v is the first Q' vertex of π, and the top Q-pocket symmetrically; if π never touches Q' there is a single Q-pocket. Since p_b and p_t are P' vertices of π, no P-pocket contains a bridge. A sight consists of a *P-part* from p to the point where it meets π, lying in one P-pocket, and a *Q-part* from there to q, lying in one Q-pocket; one part is empty when p or q is a path vertex.

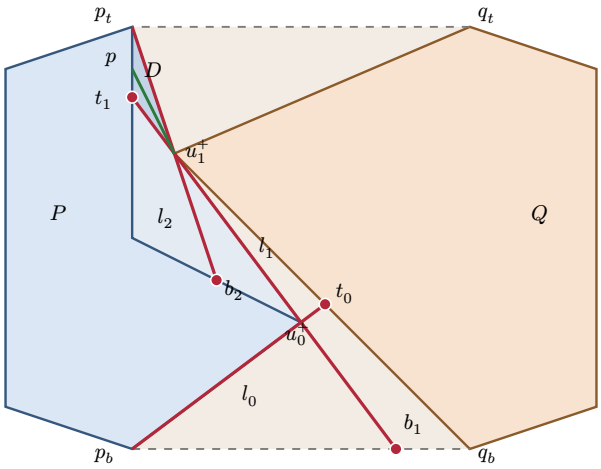


Figure A.1: Pockets and arms. The geodesic (dotted) turns left at the P' vertex $u_0^+ = (4, 3)$ and right at the Q' vertex $u_1^+ = (1, 7)$, cutting the channel into a P-pocket (blue) and two Q-pockets (sand). The two arms at u_0^+ enter the bottom Q-pocket and land on Q' (t_0) and on the bottom bridge (b_1); the two arms at u_1^+ enter the P-pocket and land on P' (t_1, b_2). The upper arm of l_1 cuts the region D (dark) off the P-pocket. The green sight from $p = (0, 9)$ to $q = u_1^+$ touches l_1 only at q: it lies above the cut t_1 of subproblem 1 and is covered by subproblem 2, whose subchains both run to their sentinels.

Lemma A4 (arms and landings). *Let u be a vertex of π at which a run ends, other than p_b or p_t. Both arms leaving u – the upper arm of the extended segment whose run ends at u and the lower arm of the one whose run begins there – lie in the pocket adjacent to u on the wall opposite to u's, and land on that pocket's arc of the opposite wall or, if $u \in P'$ and the pocket is the bottom or top Q-pocket, on a bridge. Consequently, for every i: $t_i \in P'$ if and only if $u_i^+ \in Q'$ or $i = m - 1$; t_i lies on Q' or on a bridge if and only if $u_i^+ \in P' \setminus \{p_t\}$; and symmetrically for b_i and u_i^-.*

Proof. A run ends where π turns; at an interior vertex u of π the turn wraps the polygon owning u, so by tautness the angle between the incoming and outgoing segments measured through R is at least 180°, and both prolongations – of the incoming segment beyond u and of the outgoing segment backwards beyond u – enter the interior of R on the side of π away from that polygon. An arm is an open segment in the interior of R that avoids π (Lemma A2), so it lies in a single component of $\mathrm{cl}(R) \setminus \pi$, the pocket adjacent to u on that side, and its endpoint lies on the part of ∂R in the closure of that pocket other than π: the pocket's arc of the opposite wall, and the bridges for the two end Q-pockets (Figure A.1). It cannot end at a path vertex, again by Lemma A2. The two equivalences follow, with $t_{m-1} = p_t$ and $b_0 = p_b$ unextended because the contacts are hull vertices and hence convex in R. □

Bridge landings, which violate property (iv) of [2] (Section 3.1), therefore arise only from P' vertices adjacent to an end Q-pocket. The truncation conditions of Section 4 are, by Lemma A4, conditions on the wall of the run's end vertex: the P-subchain of separator i ends at its own cut exactly when the run of l_i ends on Q' (or $i = m - 1$), and begins at the cut of $i - 1$ exactly when the run of l_{i-1} begins on Q'.

Lemma A5 (the kept sequence). *(a) $l_i \cap l_{i+1} \neq \varnothing$, and $l_i \cap l_j = \varnothing$ for $j > i + 1$: properties (i) and (ii) of [2] hold. (b) Along l_i the points $b_i, x_{i-1}, u_i^-, u_i^+, x_i, t_i$ occur in this order (with $x_{-1} = p_b$ and $x_{m-1} = p_t$). (c) If $w(i + 1) > w(i) + 1$, then u_i^+ and u_{i+1}^- lie on the same wall, the path between them is a stretch of the boundary path of a single pocket X on the opposite side, x_i lies on the upper arm of l_i and the lower arm of l_{i+1}, and the region T_i bounded by $\pi[u_i^+, u_{i+1}^-]$, $[u_i^+, x_i]$ and $[x_i, u_{i+1}^-]$ is a convex polygon whose interior lies in X and whose boundary meets the walls only at path vertices.*

Proof. (a) Step 1(d) keeps $l_{w(i+1)}$ as the largest-indexed extended segment meeting $l_{w(i)}$; every kept index beyond $w(i + 1)$ is larger, so its segment misses $l_{w(i)}$. Consecutive extended segments share a path vertex, so the sequence is well defined.

(b), (c) If the runs of l_i and l_{i+1} are adjacent, $x_i = u_i^+ = u_{i+1}^-$ by Lemma A2. Otherwise x_i lies on an arm of each. Suppose it lay on the lower arm of l_i, which sits in the pocket Y adjacent to u_i^-; the arm of l_{i+1} through x_i then also sits in Y and leaves from a vertex of Y's boundary path later than u_i^-, hence after u_i^+ (at u_i^+ itself the runs would be adjacent). If u_i^+ is on the wall opposite u_i^- it is the last vertex of Y's boundary path and no such vertex exists. If u_i^+ is on the same wall, l_i is a chord of Y through its run and the arm of l_{i+1} starts in the part of Y bounded by the upper arm of l_i, the path above u_i^+ and the wall arc above t_i; a straight segment leaving that part crosses the line of l_i once, at the upper arm, and cannot reach the lower arm. So x_i is on the upper arm of l_i, and by the mirror argument on the lower arm of l_{i+1}; this is (b). Both arms lie in the pocket X adjacent to u_i^+ on the opposite side, and u_{i+1}^- is a vertex of X's boundary path after u_i^+, so the skipped vertices between them are interior vertices of that boundary path and lie on the wall of u_i^+. The boundary of T_i, traversed from u_i^+ along the skipped chain to u_{i+1}^-, then to x_i and back, turns towards the wall at every skipped vertex (the path wraps the polygon there) and turns the same way at u_{i+1}^-, x_i and u_i^+, because the two closing segments prolong the end segments of the chain; a closed polygonal curve with all turns in one sense and total turning 2π is convex (Figure A.2). □

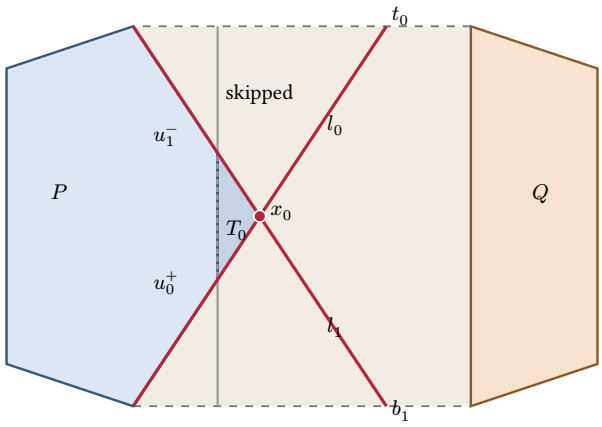


Figure A.2: A skip. The geodesic turns left twice on P'; the extension of its first segment (l_0) meets the extension of its third (l_1 in the kept sequence) at x_0, so Step 1(d) skips the middle segment's extension (grey). The skipped chain is the single P' edge from u_0^+ to u_1^-, and the region T_0 (dark) it bounds with the two arm pieces is a convex triangle. Both kept separators land on bridges, so every subchain but P_1 runs to a sentinel; P_1 starts at $p_0^+ = u_0^+$.

The greedy of Step 1(d) therefore does establish property (ii); what it does not establish is (iv) and (v). By (a) the pieces $[p_b, x_0], [x_0, x_1], \ldots, [x_{m-2}, p_t]$ form a simple arc, the *spine* of $S(P, Q)$, and $U_{<i} = l_0 \cup \ldots \cup l_{i-1}$ is connected and contains p_b.

Lemma A6 (cuts). *$l_i \cap P' \neq \varnothing$ for every i, and $p_i^+ = t_i$ if $t_i \in P'$, while $p_i^+ = u_i^+$ otherwise, in which case u_i^+ is a vertex of P'. If $t_i \in P'$ and $b_i \in Q'$, then u_i^+ is the only point of $l_i \cap Q'$ other than b_i, and b_i lies below u_i^+ in arc order on Q'. For Q: $q_i^+ = t_i$ if $t_i \in Q'$; otherwise $q_i^+ = u_i^+$ if $u_i^+ \in Q'$; otherwise $l_i \cap Q'$ is contained in $\{b_i, u_i^-\}$ and may be empty (Section 3.1), the top of l_i then lying on P' or a bridge.*

Proof. The points of $l_i \cap \partial R$ are b_i, t_i, the run vertices and any edge the run lies along, since arms are interior. If $t_i \notin P'$ then $u_i^+ \in P'$ by Lemma A4, and u_i^+ is the highest point of P' on l_i because the upper arm is interior. If $t_i \in P'$ it is the highest point of l_i altogether. For the last claim, $b_i \in Q'$ means $u_i^- \in P'$, the lower arm lands in the Q-pocket adjacent to u_i^-, whose last Q' vertex is u_i^+ (the run from u_i^- reaches Q' first at u_i^+), so b_i lies on that pocket's arc, below u_i^+. The Q statement differs from the P one only because an arm leaving a P' vertex may land on a bridge. □

Lemma A7 (first separator). *Every sight meets some l_i. Let i^* be the least such index. Then p and q lie in the same component of $\partial R \setminus U_{<i^*}$.*

Proof. $U = l_0 \cup \ldots \cup l_{m-1}$ is connected by Lemma A5(a) and contains $p_b \in l_0$ and $p_t \in l_{m-1}$; the separation principle gives the first claim, and applied to $K = U_{<i^*}$, which the sight avoids, the second. □

A.3 The coverage theorem

Throughout this section (p, q) is a visible pair other than the two bridge pairs, and $i = i^*$ is the index of Lemma A7. We treat the P side; the Q side is the mirror image.

Lemma A8 (start side). *If $i > 0$ and $b_{i-1} \in P'$, then p lies strictly above p^+_{i-1} in arc order.*

Proof. By Lemma A4, $u^-_{i-1} \in Q'$, so the lower arm of l_{i-1} lies in the P-pocket Π' whose boundary path contains u^-_{i-1}, and b_{i-1} is on the arc of Π'. The sight avoids l_{i-1}, so $p \neq b_{i-1}$ and $p \neq p^+_{i-1}$.

If p is below b_{i-1}: both p_b and b_{i-1} belong to $U_{<i}$, so the component of $\partial R \setminus U_{<i}$ containing p is a sub-arc of $P'(p_b, b_{i-1})$ and contains no point of Q', contradicting Lemma A7.

If $b_{i-1} < p < p^+_{i-1}$: by Lemma A6, p^+_{i-1} is u^+_{i-1} (when that vertex is on P'; it is then the top vertex of Π', being the first P' vertex after u^-_{i-1}) or t_{i-1} (when $u^+_{i-1} \in Q'$, so that the upper arm also lies in Π'). In either case $l_{i-1} \cap \mathrm{cl}(\Pi')$ is a chord of Π' from b_{i-1} to p^+_{i-1} through the run, and the region D' bounded by $P'[b_{i-1}, p^+_{i-1}]$ and this chord meets Q' only at run vertices, which lie on l_{i-1}. The open sight starts inside D', and $q \notin D'$ since a point of Q' in D' would lie on l_{i-1}; so the sight leaves D' through P' (impossible by visibility) or through the chord (impossible by the choice of i). In the mirror statement for Q, when l_{i-1} meets Q' only at b_{i-1} the cut is b_{i-1} itself and this second case is empty. □

Lemma A9 (end side). *Suppose $t_i \in P'$, so that $e^P_i = p^+_i = t_i$. Then either p lies at or below t_i in arc order, or $[p, q] \cap l_i = \{u^+_i\}$ and $q = u^+_i$.*

Proof. If $i = m - 1$ then $t_i = p_t$ and there is nothing to prove, so let $i < m - 1$; by Lemma A4, $u^+_i \in Q'$. Let $\Pi'' = \Pi(v_a, v_c)$ be the P-pocket whose boundary path contains u^+_i. The upper arm $A^+ = (u^+_i, t_i]$ is a chord of Π'' landing on $P'(v_a, v_c)$; it cuts off the region D bounded by $P'[t_i, v_c]$, A^+ and $\pi[u^+_i, v_c]$. The rest of Π'' has, as its boundary path, $\pi[v_a, u^+_i]$, whose last segment is the run of l_i; if $u^-_i \in Q'$ the lower arm of l_i lies in it too and $l_i \cap \mathrm{cl}(\Pi'')$ is the whole chord $[b_i, t_i]$. Suppose p is above t_i.

Case $p \in P'(t_i, v_c)$. The P-part of the sight starts in D and stays in $\mathrm{cl}(\Pi'')$ until it meets π. It cannot cross A^+: the region beyond has boundary P' together with pieces of l_i only (the run, or the whole chord), and its points of Q' lie on l_i; having crossed the line of l_i once the sight can meet it no more, so it could neither leave that region nor end in it. It cannot touch A^+ without crossing, since that forces collinearity with l_i and puts t_i in the open sight. If it leaves D through $\pi(u^+_i, v_c)$ it enters a Q-pocket whose boundary path lies at or above u^+_i, whereas the remaining pieces of l_i – the run, below u^+_i on π, and the lower arm, which lies in Π'' below A^+ or in the Q-pocket whose boundary path ends at u^+_i – are out of reach, contradicting $[p, q] \cap l_i \neq \varnothing$. What remains is that the sight ends at $q = u^+_i$ and meets l_i nowhere else (the green sight of Figure A.1, with $i = 1$).

Case $p \geq v_c$. The P-part lies in a pocket at or above v_c and the Q-part in a Q-pocket whose boundary path lies at or above v_c, so no arm of l_i is reachable, and a point of l_i on π can be reached only along π, the sight meeting π in a connected set; a sight containing a path vertex in its interior violates visibility, so the only possibility is $p = v_c$ with the sight equal to the path segment $[u^+_i, v_c]$, which is again $q = u^+_i$ and a single touch. □

The exceptional alternative of Lemma A9 is the configuration of Section 3.4, seen from the other wall: the sight leaves the top vertex of the run of l_i into the region above l_i's upper arm, and the cut t_i, which lies below p in arc order, shields nothing. This is precisely the step "no point above q^+_{i+1} is visible from l_i" in the proof of Lemma 1 of [1], whose justification is property (v). The next lemma shows that the neighbouring subproblem catches such a pair.

Lemma A10 (witness transfer). *Suppose $t_i \in P'$, $q = u^+_i$ and $[p, q] \cap l_i = \{q\}$. Then $p \in P_{i+1}$ and $q \in Q_{i+1}$.*

Proof. $q \in Q_{i+1}$: the start of Q_{i+1} is q_b unless $b_i \in Q'$, in which case it is $q_i^+ = u_i^+ = q$ by Lemma A6. The end is q_t unless $t_{i+1} \in Q'$; then $u_{i+1}^+ \in P'$ and t_{i+1} lands on the arc of the Q-pocket adjacent to u_{i+1}^+, whose first Q' vertex is at or after u_i^+ in path order, hence at or above q in arc order by Lemma A3.

$p \in P_{i+1}$: the start is p_b or $p_i^+ = t_i$, both below p. The end is p_t unless $t_{i+1} \in P'$, i.e. $u_{i+1}^+ \in Q'$. Then u_{i+1}^+ is a vertex of the boundary path of Π'' after u_i^+, and the upper arm of l_{i+1} is a chord of the region D of Lemma A9 from u_{i+1}^+ to $t_{i+1} \in P'(t_i, v_c)$, so $t_{i+1} > t_i$. We show $p \le t_{i+1}$. If the runs of l_i and l_{i+1} are adjacent, l_{i+1} passes through q; the sight ends at q and so meets the line of l_{i+1} only there, hence cannot cross that arm, and p lies on the near side of it. Otherwise (Lemma A5(c)) the sight approaches q through the corner of the convex region T_i at q, and since it does not cross $[q, x_i] \subset l_i$ it must enter T_i across $(x_i, u_{i+1}^-) \subset l_{i+1}$; a second crossing of the line of l_{i+1} at the upper arm is impossible, so again $p \le t_{i+1}$. In the situation $p = v_c$ of Lemma A9 the runs are adjacent, since v_c and u_i^+ lie on different walls. □

Theorem A. *Let P and Q be simple polygons with disjoint boundaries in the non-containing case, with an unpinched channel, and let $S(P, Q)$ and the subchains be as in Section 4. For every visible pair (p, q) there is an index i with $p \in P_i$ and $q \in Q_i$. Specifically, with i^* as in Lemma A7, either $(p, q) \in P_{i^*} \times Q_{i^*}$, or $[p, q] \cap l_{i^*} = \{u_{i^*}^+\}$ with $u_{i^*}^+ \in \{p, q\}$ and $(p, q) \in P_{i^*+1} \times Q_{i^*+1}$.*

Proof. The bridge pairs are covered by subproblems 0 and $m-1$: both subchains of subproblem 0 start at their sentinels, and those of $m-1$ end at $p_{m-1}^+ = p_t$ and at the sentinel q_t ($t_{m-1} = p_t \notin Q'$). For any other pair let $i = i^*$. On the P side, p is above s_i^P by Lemma A8 (the sentinel case being trivial) and at or below e_i^P by Lemma A9 unless the exceptional alternative holds; symmetrically on the Q side. The two exceptional alternatives are exclusive, u_i^+ lying on one wall, and in either Lemma A10 or its mirror image places both p and q in subproblem $i+1$; note $i < m-1$ there, since $t_{m-1} = p_t$ never has p above it. Hence $\sigma(P, Q) = \min_i \sigma(P_i, Q_i)$. □

A.4 Remarks

The wrap override. The proof never invokes the whole-boundary override of Section 3.4: with the cuts identified by Lemma A6 and the slices of Lemma A8, Lemma A9 and Lemma A10, the anchor cuts x_{i-1}^+ and x_i^+ always lie inside the slice whenever the corresponding truncation applies, and the pair is covered without it. The configuration that triggers it – a later separator landing below an earlier one's cut – does occur (about one pair in ten thousand in the harness of Section 6), but the pairs it concerns are covered by the neighbouring subproblems. Our implementation no longer applies it.

Which cut. The proof uses the paper's cut, the highest point of $l_i \cap X'$ along l_i. By Lemma A6 this is t_i or the top run vertex, and in the one place where two orders could disagree (Lemma A6, last claim) they agree. Redefining the cut as the last point of $l_i \cap X'$ in arc order changes the subchains in about a tenth of random instances but no coverage outcome; we tested it and did not adopt it.

What fails in the original. Every step of the proof of Lemma 1 of [1] survives except the one that needs property (v): with the minimal index in place of the hypothesis $pq \cap l_{i-1} = \varnothing$, Lemma A8 is the claim that p and q are above the previous cuts, and Lemma A9 is the claim that they are below the current ones. The exceptional alternative is where a grazing or wrongly landing separator leaves a visible pair above its own cut; the paper's remedy is to end the subchain at the next separator's cut, which is sound only under property (v), and the sentinel is the replacement that needs no such property. The bridge landings of Section 3.1 enter only through Lemma A4: a separator whose run ends on P' lands on Q' or a bridge, and in both cases its Q-subchain runs to the sentinel.

Complexity. By Lemma A4 the run of l_i ends on one wall, so at the top end of subproblem i exactly one subchain is truncated, the one on the other wall, and the other runs to its sentinel; likewise at the bottom end, according to the wall on which the run of l_{i-1} begins; a bridge landing leaves both at the sentinel. Every subproblem therefore contains, at each end, the whole remainder of one facing arc. A channel whose geodesic alternates between the walls $\Theta(n)$ times has $\Theta(n)$ subproblems each containing a whole facing

arc, so the aggregate subproblem size is $\Theta(n^2)$: the bound stated in Section 4 is tight, and it is the typical case. Nor need a sentinel-extended subchain lie on one side of the line of l_i (both subchains of subproblem 2 in Figure A.1 are whole arcs), so such subproblems are not linearly separable; an implementation that solves them by a quadratic method runs in $O(n^3)$ overall, slower than the brute force it set out to improve on. Bounding the sentinel side by a later cut is exactly what property (v) supplies, and the two-path construction of [2] remains the route to it.

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